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CHIP PACKAGE ASSEMBLY WITH ON-PACKAGE CONTAINMENT SYSTEM

Abstract

A chip package, an electronic device, and methods for fabricating the same are disclosed herein. In one example, a chip package includes a package substrate, an integrated circuit (IC) die, and a stiffener. The IC die has a first height and is mounted on a top surface of the package substrate. The stiffener is mounted to the top surface of the package substrate. A cavity is disposed in the stiffener. The cavity has an opening formed through the stiffener that faces towards the IC die.

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Background/Summary

TECHNICAL FIELD

[0001] Embodiments of the present disclosure generally relate to chip package assemblies and techniques for manufacturing the same. In particular, a chip package assembly is described herein having features that mitigate thermal interface materials (TIM) from leaking out of the contained areas.

BACKGROUND ART

[0002] Electronic devices, such as tablets, computers, server, in-door telecom, out-door telecom, industrial computers, high performance computing data centers, copiers, digital cameras, smart phones, control systems and automated teller machines, among others, often employ electronic components which leverage chip package assemblies for increased functionality and higher component density. Conventional chip package assemblies include one or more stacked components such as integrated circuit (IC) dies, through-silicon-via (TSV) interposer, and a package substrate, with the chip package assembly itself stacked on a printed circuit board (PCB). The IC dies may include memory, logic, MEMS, RF or other IC devices.

[0003] Surface mounted components, such as capacitors, resistors, and the like, are often utilized in chip package assemblies in between the IC dies and the sides of the package. The surface mounted components are typically mounted to a package substrate laterally to the side and outward of the dies that are mounted to the package substrate. The surface mounted components are typically mounted between the dies and a stiffener, if utilized, which is generally mounted at the periphery of the of the package substrate. In some cases, a lid will be attached along the edges of the package substrate enclosing the IC dies, surface mounted components and, stiffener. Electronic devices, such as those listed above, produce heat and often require a heat sink to re-distribute the heat to allow for proper operating conditions for the electronic devices.

[0004] Often a thermal interface material (TIM) is utilized between the IC die and heat sink to provide a path for conducting the heat from the electronic device to the heat sink. The TIM may flow out from between the die and the heat sink into the region containing the surface mounted components between the IC die and the stiffener ring, which can short the components, and often the TIM overflows over the stiffener ring, leaking outside of the package. Each is considered a production defect and can adversely affect the electronic device or other nearby electronic devices and components. Thus, to minimize the potential for production defects due to TIM leaking from the space between the IC die and the heat sink, the space for surface mounted components must be sufficiently spaced from both the dies and the stiffener, or a foam or gasket must be applied around the outside of the package after assembly, which undesirably reduces the space available for dies or undesirably requires a larger package substrate, or in some instances an insufficient amount of TIM may be applied to the package.

[0005] Therefore, a need exists for an improved chip package assembly that mitigates leaking of excess TIM as compared what is conventionally available in the art.

SUMMARY

[0006] A chip package, an electronic device, and methods for fabricating the same are disclosed herein. In one example, a chip package includes a package substrate, an integrated circuit (IC) die, and a stiffener. The IC die has a first height and is mounted on a top surface of the package substrate. The stiffener is mounted to the top surface of the package substrate. A cavity is disposed in the stiffener. The cavity has an opening formed through the stiffener that faces towards the IC

die.

[0007] In another example, an electronic device is provided. The electronic device includes a package substrate, an integrated circuit (IC) die, a first electronic component, a heat sink, and a stiffener. The package substrate has a top surface. The IC die is mounted on the top surface of the package substrate. The stiffener is also mounted to the top surface of the package substrate. A first electronic component is mounted to a first region of the package substrate defined between the stiffener and the IC die. A cavity is formed in the stiffener. The cavity has an opening formed through the stiffener facing the first IC die. The heat sink is mounted on the first IC die. Thermal interface material (TIM) is disposed between the first IC die and the heat sink.

[0008] In yet another example, a method for fabricating a chip package assembly is provided that includes mounting an IC die to a top surface of a package substrate; attaching a stiffener to the top surface of the package substrate, the stiffener having a cavity facing the IC die; applying thermal interface material (TIM) to one of the top of the IC die or a bottom of a heat sink; and mounting the heat sink on the top of the IC die sandwiching the TIM therebetween.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0009] So that the manner in which the above recited features of the present invention can be understood in detail, a more particular description of the invention, briefly summarized above, may be had by reference to embodiments, some of which are illustrated in the appended drawings. It is to be noted, however, that the appended drawings illustrate only typical embodiments of this invention and are therefore not to be considered limiting of its scope, for the invention may admit to other equally effective embodiments.

[0010] FIG. 1 is a cross sectional schematic view of an integrated circuit chip package mounted on a printed circuit board.

[0011] FIG. 2 is a partial schematic view of the integrated circuit chip package of FIG. 1.

[0012] FIG. 3 is a partial schematic view of a portion of another integrated circuit chip package.

[0013] FIG. 4 is a schematic top view of an example of a package substrate.

[0014] FIG. 5 is a flow diagram of a method for forming a chip package.

[0015] To facilitate understanding, identical reference numerals have been used, where possible, to designate identical elements that are common to the figures. It is contemplated that elements of one example may be beneficially incorporated in other examples.

DETAILED DESCRIPTION

[0016] Embodiments of the disclosure generally provide chip package assemblies and package substrates that mitigate the potential for thermal interface materials (TIM) from leaking out from the area between the IC die and heat sink (bond line), contacting surface mounted components and/or overflowing over the stiffener ring. The chip package assemblies package substrates and techniques for manufacturing the same advantageously increase product yield by utilizing a containment structure within the stiffener ring to essentially prevent TIM from overflowing over the stiffener ring. Moreover, reliability is improved by insuring TIM does not contact the surface area outside of the package (i.e. outside of the stiffener), which reduces the potential for issues to other electronic devices and components outside of the electronic device on a printed circuit board (PCB). Other advantages include a more local containment area to the desired bond line, an economic solution avoiding the need for a separate containment fence, such as a foam or gasket applied outside the boundary of the stiffener, and improved package warpage resistance as the containment system is contained within a stiffener, all without increase in overall package substrate size.

[0017] Turning now to FIG. 1, a cross section of an integrated circuit electronic device **110** is

illustrated having an integrated circuit (IC) chip package assembly **100** mounted on a printed circuit board (PCB) **112** by solder balls **134**. The PCB **112** and solder balls **134** are shown in phantom. The chip package assembly **100** generally includes one or more IC dies **102**, a package substrate **106**, one or more surface mounted components **104**, and a stiffener **122**. The stiffener **122** is configured to contain the movement of thermal interface material (TIM) **116**, as further described below. The IC dies **102** may be programmable logic devices, such as field programmable gate arrays (FPGA), memory devices, optical devices, processors or other IC logic structures. The surface mounted components **104** may be one or more of a capacitors, diodes, resistors, inductors, or other suitable discrete circuit elements.

[0018] The IC dies **102** are connected directly to the package substrate **106**, for example as flip chip ball grid array (FCBGA), ball grid array (BGA), wire bond and the like. In another example, one or more IC dies **102** of the chip package assembly **100** may be connected to the package substrate **106** via an interposer (not shown). It is contemplated that the chip package assembly **100** may have other configurations. Although one IC die **102** is shown in FIG. **1**, the number of IC dies may range from one to as many as can be fit within the chip package assembly **100** in order to meet design criteria. A bottom of the IC die **102** is connected to a top surface **114** of the package substrate **106** by solder connections **108**. The solder connections **108**, such as micro-bumps, mechanically and electrically connect the circuitry of the IC die **102** to the circuitry of the package substrate **106**.

[0019] The electronic device **110** includes a heat sink **115** positioned over the IC die **102**. The heat sink **115** may be an active or a passive heat sink device. The heat sink **115** is used to dissipate the heat from an IC die, such as the IC die **102**, such that the temperature of the IC die **102**, the surface mounted components **104**, and any other electronic devices mounted on the PCB **112** nearby to electronic device **110**, can be maintained in the operating range of each component or device. The heat sink **115** may be attached to the PCB **112** via an interposer (not shown) or directly to the PCB **112**.

[0020] The TIM **116** is disposed in the interstitial spaces between the bottom **119** of the heat sink **115** and the top surface **120** of the IC die **102**. TIM **116** provides a path with increased conductivity to transfer the heat produced by a heat producing device, such as IC die **102**, to a heat sink, such as heat sink **115**, whereby the heat sink is configured to more efficiently dissipate heat. In some examples, the TIM **116** may be a liquid metal TIM. In other examples, the TIM **116** may be a thermal paste, a thermally conductive putty, or other types of TIM.

[0021] A TIM-containing stiffener **122** is generally mounted at an edge **136** along the periphery of the package substrate **106**. The stiffener **122** often surrounds the IC die **102** and surface mounted components **104** continuously circumscribed along the outer edge of the chip package assembly **100**. In the example illustrated in FIG. **1**, an inner side wall **130** of the stiffener **122** faces and is spaced a distance apart from the IC die **102**. An outer side wall **160** of the stiffener **122** faces away from the IC die **102** and forms the outer perimeter of the stiffener **122**.

[0022] The stiffener **122** is attached to the top surface **114** of the package substrate **106**. In some examples, the stiffener **122** is attached to the package substrate **106** by an adhesive material, such as an epoxy or other suitable bonding material. The stiffener **122** provides mechanical support which helps prevent the chip package assembly **100** from bowing and warping. The stiffener **122** may be a single layer structure or a multi-layer structure. The stiffener **122** may be made of ceramic, metal or other various inorganic materials, such as aluminum oxide (Al.sub.2O.sub.3), aluminum nitride (AlN), silicon nitride (SiN), silicon (Si), copper (Cu), aluminum (Al), and stainless steel, among other materials. The stiffener **122** can also be made of organic materials such as copper-clad laminate.

[0023] The surface mounted components **104** are coupled to the top surface **114** of the package substrate **106** in a region **132** defined between the inner side wall **130** of the stiffener **122** and the side **118** of the IC die **102**. The surface mounted components **104** may be soldered to a conductor

residing on or in the package substrate **106**, and are coupled to the circuitry of the die **102** through the circuitry of the package substrate **106**.

[0024] The TIM **116** is often a viscous material, such as a liquid metal or a paste, which may leak out from the area below the heat sink **115** and inwardly bounded by the sides **118** of the IC die **102**. The TIM **116** may thus flow into the region **132** between the IC die **102** and the stiffener **122**. The TIM **116** may be an electrically conductive material which can short the surface mounted components **104**. To prevent shorting, the surface mounted components **104** are covered by an encapsulant **105**. The encapsulant **105** protects the materials and underlying components (for example, the surface mounted components **104**) from oxidation, moisture, as well as from unintended electrical connections or shorting. The encapsulant **105** may be formed from dielectric materials such as epoxy, resin, or other polymers.

[0025] In conventional designs, an excess amount of the TIM **116** may flow into the region **132** filling the available space and overflowing over the top of conventional stiffeners. The TIM **116** may flow out of conventional chip package assemblies and onto the PCB, thereby potentially affecting the electrical connections made by the solder balls, or any other chip packages, components, or other electronic devices on the PCB.

[0026] To mitigate the undesirable movement of TIM in conventional chip package designs, the stiffener **122** includes a cavity **126** disposed within the stiffener **122**. The cavity **126** has an opening **127** on the inner side wall **130** of the stiffener facing toward the IC die **102**. The cavity **126** does not include an opening that penetrates the outer side wall **160** of the stiffener **122**. The opening **127** allows the flow of TIM **116** to enter the cavity **126** and fill up the space of the cavity **126** instead of flowing over the top **128** of the stiffener **122**, as would undesirably happen in conventional chip packages. Thus, the cavity **126** substantially prevents TIM **116** from leaking over the stiffener **122** and extending beyond the peripheral edge of the package substrate **106**, by providing a space for the excess TIM **116** to go, which may inadvertently prevent the chip package assembly **100** from being utilized or otherwise disrupt the function of the electrical device **110** or other devices on the PCB **112**. As the cavity **126** does not include an opening that penetrates the outer side wall **160** of the stiffener **122**, TIM **116** within the cavity **126** cannot flow through the stiffener **122**.

[0027] FIG. 2 is a partial sectional view of the chip package assembly **100** illustrating the stiffener **122** having the internal cavity **126** facing the region **132** with surface mount components **104** disposed next to the IC die **102** and the heat sink **115** in greater detail. The stiffener **122** may be configured in various embodiments having different sizes and shapes of the cavity **126**, and the examples shown herein are not intended to limit the cavity **126** to a particular size or shape.

[0028] The opening **127** is formed in the inner side wall **130** of the stiffener **122** and provides access to the cavity **126** from the region **132**. The bottom edge of the opening **127** is disposed at a height **227** about the top surface **114** of the package substrate **106**. The portion of the inner side wall **130** disposed below the opening **127** creates a cavity in the region **132** containing the surface mount components **104**. TIM **116** entering this cavity defined in the region **132** would move up the inner side wall **130** when encountering the stiffener **122** until the TIM **116** reaches the opening **127**. Upon reaching the opening **127**, the TIM **116** would flow through the opening **127** and into the cavity **126** formed in the stiffener **122** instead of continuing up the inner side wall **130** and overflowing the top **128** of the stiffener **122**. The cavity **126** thus prevents the free flow of TIM **116** from flowing over the top **128** of the stiffener **122**. In one example, the height of the bottom of the opening **127** has a height **227** that is greater than the height **220** from the top surface **114** of the package substrate **106** to the top of the IC die **102**. In some examples, the TIM **116** is viscous and spreads quickly, thus a height **227** to the bottom of the opening **127** being greater than the height **220** of the top surface **120** of the IC die **102** ensures that the TIM **116** will first fill the space between the IC die **102** and the heat sink **115** before the TIM **116** flows into the cavity **126**, while still inhibiting the flow of TIM **116** over the top **128** of the stiffener **122**.

[0029] In another example, a side wall **230** of the cavity **126** has a length **202** defined below a

bottom of the opening **127** is long enough to define a volume of the cavity **126** sufficiently great enough to contain the potential overflow of TIM **116**, such that the TIM **116** does not flow over the top **128** of the stiffener **122**. In yet another example, the height **227** of the bottom of the opening **127** is sufficiently tall enough to cause the TIM **116** to flow upward along the inner side wall **130** of the stiffener **122** and in contact with the side **118** of the die **102** and the stiffener **122** first, while the cavity **126** provides an safeguarding cavity to substantially prevent the TIM **116** from flowing over the top **128** of the stiffener **122**. Thus, the height **227** of the opening **127** in the stiffener **122**, along with the presence of the cavity **126**, advantageously does not require the application of precise amounts of TIM **116** or other mitigation efforts to prevent TIM **116** from flowing out of the chip package assembly **100**. In other examples, the side wall **230** of the cavity **126** is defined below a top of the opening **127**, and in still other examples, a first portion of the side wall **230** of the cavity **126** is defined below a top of the opening **127** while a second portion of the side wall **230** of the cavity **126** is defined above a top of the opening **127** such that the opening **127** is in the middle of the side wall **230**.

[0030] In one example, the side wall **230** and other structure bounding the cavity **126** is a part of the stiffener **122**. This is, the side wall **230** and other structure bounding the cavity **126** is a contiguous single component with the stiffener **122**. In another example, the side wall **230** is separately formed and attached to the stiffener **122**. In yet another example, the side wall **230** and other structure bounding the cavity **126** are formed from a different material than the stiffener **122**, and separately attached to the package substrate **106**. In yet another example, the side wall **230** may be a separate component disposed or formed on the top surface **114** of the package substrate **106**.

[0031] The remaining sides of cavity **126** may be formed within the stiffener **122**. In some examples, the cavity **126** is formed prior attaching the stiffener **122** to the package substrate **106**. In other examples, the cavity **126** may be formed after the stiffener has been attached to the package substrate **106**.

[0032] FIG. **3** is a partial schematic view of a portion of another integrated chip package assembly **300**. Although only a portion of the chip package assembly **300** is illustrated in FIG. **3**, the chip package assembly **300** is substantially the same as the chip package assembly **100** depicted in FIGS. **1-2**, except wherein the heat sink **115** is configured to extend over the region **132** closer to the stiffener **122**. FIG. **3** also depicts the relationship of the height **227** of the opening **127**, the top **128** of the stiffener **122**, the top surface **120** of the IC die **102**, the bottom **119** of the heat sink **115**, and the flow of the TIM **116** out from between the IC die **102** and the heat sink **115**.

[0033] The heights **227** of the opening **127** in connection with the top **128** of the stiffener **122** and the height of the top surface **120** of the IC die **102** are selected to ensure that the TIM **116** is able to fill the gap formed between the IC die **102** and the heat sink **115**, before any displacement of TIM **116** out from under the gap is preferentially directed toward the surface mounted components **104** and then inward through the opening **127** and into the cavity **126**.

[0034] The number, location, and volume of the cavity **126** is selected to accommodate an expected amount of TIM **116** expected to be present after filling the gap between the IC die **102** and the heat sink **115**. The location of the cavity **126** may be selected to promote a desired displacement of the TIM **116**, such that the TIM **116** maintains contact with both the IC die **102** and heat sink **115**, while the cavity **126** contains the overflow of the TIM **116**, thus substantially preventing any TIM **116** from flowing over the top **128** of the stiffener. Accordingly, the cavity **126** advantageously prevents TIM **116** from being present outside of the chip package **100** or leaking onto the PCB **112**, where the TIM **116** could potentially interfere with the chip package assembly **100** or any other electronic devices disposed on the PCB **112**.

[0035] In some examples, the cross section of the cavity **126** may be rectangular. In other examples, the cross section of the cavity **126** may be round, oblong, or triangular. In some examples, the vertical cross-sectional area of the profile of the cavity **126** may greater than vertical cross-sectional area of the opening **127**.

[0036] The heat sink **115** may be configured, such as shown in FIG. 3, to have sides **315** that extend beyond the sides of the sides **118** of the IC die **102** and over the surface mounted component **104** region **132**. In some examples, the bottom **119** of the heat sink **115** may have a step **316** formed having an inverse shape of the corner of the stiffener **122**. In some examples, the step **316** includes a roof **318** that faces the same direction as the bottom **119** of the heat sink **115**. The roof **318** is generally above the top **128** of the stiffener **122**. In some examples, the roof **318** horizontally overlaps the top **128** of the stiffener **122** such that the step **316**, the roof **318** and top corner of the stiffener **122** form a tortuous gap **320**. The tortuous gap **320** discourages the flow of TIM **116** therethrough. The size of the gap **320** relative to the size of the opening **127** also promotes capillary forces created by decreasing the space between the stiffener **122** and the heat sink **115** to leverage the surface tension of the TIM **116** to direct the excess TIM flow preferentially into the cavity **126** before flowing upwards over the top **128** of the stiffener **122**.

[0037] Referring now to the schematic top view of one example of the package substrate **106** illustrated in FIG. 4, the stiffener **122** contiguously circumscribes the IC die **102** (shown in phantom below the heat sink **115**) along the boundary of the package substrate **106**. The heat sink **115** extends beyond the sides **118** of the IC die **102**. Similarly, the cavity **126** (also shown in phantom) within the stiffener **122** contiguously circumscribes the IC die **102**. The stiffener **122** and cavity **126** may be fabricated as described above, or in other manner in which stiffeners are used in package devices. In other examples, the stiffener **122** may be attached to the package substrate **106** in one or more areas along the periphery of the package substrate **106**, wherein the cavity **126** is only in the areas immediately adjacent an IC die **102**.

[0038] FIG. 5 is a flow diagram of a method **500** for forming a chip package, such as the chip package assemblies **100**, **300** described above, among others. The method **500** begins at operation **502** by mounting one or more IC dies **102** on a top surface **114** of a package substrate **106**. The IC dies **102** may be mounted to the top surface **114** of the package substrate **106** utilizing conventional or other suitable techniques, such as flip chip ball grid array (FCBGA), ball grid array (BGA), wire bond, or solder balls. In one example, a heat sink **115** is disposed over the first IC die **102**, wherein the heat sink **115** over the first IC die **102** forms a gap between the heat sink **115** and the first IC die **102** defined by a first gap.

[0039] At operation **504**, a stiffener **122**, having an inner side wall **130** and a top **128**, is attached to the top surface **114** of the package substrate **106** by an adhesive or other suitable method. In some examples, the stiffener **122** and a side of the heat sink **115** are in closer proximity than the stiffener **122** and a side of the IC die **102**.

[0040] The stiffener **122** includes one or more internal containment areas (e.g., cavities **126**) formed in the stiffener **122**. The cavity **126** has an opening **127** formed through the inner side wall **130** of the stiffener **122** and facing a region **132** between the stiffener **122** and the IC die **102**.

[0041] At operation **506**, TIM **116** is applied to one of a top of the first IC die **102** or a bottom of a heat sink **115**. At operation **508**, the heat sink **115** is mounting on the top of the first IC die **102**, sandwiching the TIM **116** therebetween. In some examples, some of the TIM **116** at operation **508** is squeezed out from between the first IC die **102** and the heat sink **115**. In some examples, the portion of the TIM **116** squeezed out from between the first IC die **102** and the heat sink **115** extends into the region **132** between the stiffener **122** and the IC die **102** and enters the cavity **126** disposed in the stiffener **122** through the opening **127** of the cavity **126**, thus substantially preventing the flow of TIM **116** over the top **128** of the stiffener **122**.

[0042] Thus, chip package assemblies and package substrates that mitigate the potential for adhesive and/or underfill from contacting surface mounted components, along with techniques for fabricating the same, have been described above. The chip package assemblies, package substrates and techniques for manufacturing the same advantageously increase product yield, provide improved reliability by insuring consistent surface contact area between the package substrate and a stiffener, thus providing greater delamination resistance. Additionally, signal integrity is improved

due to the closer spacing between surface mounted components and dies. Furthermore, the embodiments described above allow for smaller package substrate area requirements, and improved package warpage resistance as wider stiffeners may be accommodated due to smaller stiffener to die spacing requirements without an increase in package substrate size.

Claims

1. A chip package comprising: a package substrate having a top surface; an integrated circuit (IC) die having a first height mounted on the top surface of the package substrate; a stiffener mounted to the top surface of the package substrate, the stiffener having a first side facing the IC die and a top; and a cavity disposed in the stiffener, the cavity having an opening formed through the first side of the stiffener.
2. The chip package of claim 1, wherein the opening in the stiffener is on the first side of the stiffener and at a second height that is greater than the first height of the IC die.
3. The chip package of claim 1, wherein the stiffener is positioned at an outer border of the package substrate.
4. The chip package of claim 1, wherein a heat sink is disposed over the IC die.
5. The chip package of claim 4, wherein a space between the heat sink and the stiffener forms a tortuous gap.
6. The chip package of claim 5 further comprising: a thermal interface material (TIM) is disposed between the IC die and the heat sink, the TIM extending preferentially into the cavity in the stiffener relative the tortuous gap.
7. The chip package of claim 5, wherein a distance across the opening is larger than a distance across the tortuous gap.
8. The chip package of claim 1, wherein the cavity extends away from the top of the stiffener.
9. An electronic device, comprising: a package substrate having a top surface; an integrated circuit (IC) die having a first height mounted on the top surface of the package substrate; a stiffener mounted to the top surface of the package substrate, the stiffener having a first side and a top; a first electronic component mounted to a first region of the package substrate defined between the stiffener and the IC die; a cavity formed in the stiffener, the cavity having an opening formed through the stiffener facing the IC die; a heat sink mounted on the IC die; and a thermal interface material (TIM) disposed between the IC die and the heat sink.
10. The electronic device of claim 9, wherein the opening in the stiffener is on the first side of the stiffener and at a second height that is greater than the first height of the IC die.
11. The electronic device of claim 9, wherein the stiffener is positioned at an outer border of the package substrate.
12. The electronic device of claim 9, wherein a tortuous gap is defined between the heat sink and the stiffener.
13. The electronic device of claim 12 wherein the TIM extends preferentially into the cavity in the stiffener relative the tortuous gap.
14. The electronic device of claim 9, wherein the TIM extends into the cavity of the stiffener.
15. The electronic device of claim 9, wherein the cavity extends away from the top of the stiffener.
16. A method for fabricating a chip package assembly, the method comprising: mounting an IC die to a top surface of a package substrate; attaching a stiffener to the top surface of the package substrate, the stiffener having a cavity facing the IC die; applying thermal interface material (TIM) to one of the top of the IC die or a bottom of a heat sink; and mounting the heat sink on the top of the IC die sandwiching the TIM therebetween.
17. The method of claim 16 further comprising flowing the TIM into the cavity.
18. The method of claim 16, wherein mounting the heat sink on the top of the IC die further comprises: forming a tortuous path between the heat sink and the stiffener.

19. The method of claim 18 further comprising preferentially flowing the TIM into the cavity relative to the tortuous path.

20. The method of claim 17, wherein flowing the TIM into the cavity further comprises flowing the TIM downward within the cavity.
